Research Summary

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Catalysts work under potential, solvent, adsorbates and electric fields that reshape their structure and electrostatics, yet most models treat them as static and unpolarized. My research asks **how these operating conditions change a catalyst's structure and reactivity**. To answer that, I build simulation, machine-learning and AI tools.

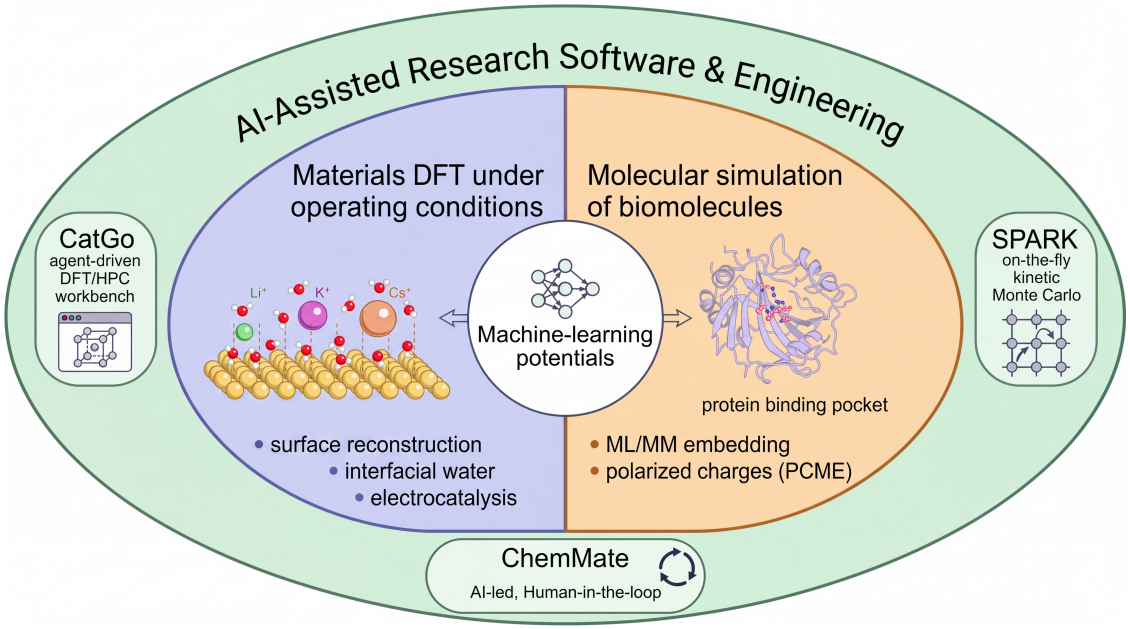


Figure 1. Two research interests linked by machine-learning interatomic potentials, built on AI-assisted research infrastructure.

(a) Surface reconstruction and interfacial structure *What does a working interface look like?*
I started my research with the interfacial solvent: ab initio MD showed that surface defects anchor interfacial water and raise its cavitation free energy, which controls how hydrophobic molecules reach the surface (*Chem. Sci.* **2026**, *17*, 15735–15744). The same water structure explained a puzzle in cation effects: Au surface geometry templates interfacial water enough to reverse the cation trend in CO_2 reduction (*Angew. Chem. Int. Ed.*, under review). Turning from the solvent to the catalyst itself, I used constant-potential ensembles to show that the size effect of IrO_x clusters in acidic OER is set by the populations of local Ir–oxo environments (*Adv. Funct. Mater.*, under review).

At UCLA, I now study Cu surfaces that restructure under reaction, using site-resolved KMC simulations to find which kinetically trapped sites dominate the C–C coupling rate.

(b) Catalytic mechanisms and design *From active site to overall activity*
At Fudan University, I tested a Zn–O–Fe photocatalyst and used DFT to show that the medium Fe spin state selects $\bullet CH_2OH$ over $\bullet OCH_3$, which steers CH_4 oxidation toward ethanol (*Angew. Chem. Int. Ed.* **2025**, *64*, e202510241). I then moved from mechanism to design. With machine learning (a stability-first screen and an interpretable SISSO descriptor), I ranked doped MoS_2 for nitrogen reduction over HER (ChemRxiv 2026).

(c) AI-assisted scientific software *Making these campaigns possible in the AI era*
The ensembles and screens above meant thousands of calculations across computing clusters, so I developed tools to speed them up. CatGo lets AI coding agents drive an interactive structure/DFT/HPC workbench, and every action can be reviewed (*Digit. Discov.* **2026**, *5*, 3825–3849). ChemMate is a human-in-the-loop infrastructure for AI-assisted research across the whole research process: AI carries out the steps, and researchers make the decisions (ChemRxiv 2026).

(d) Molecular modeling: electrostatics in enzyme design *Beyond heterogeneous catalysis*
The environment matters just as much in biological systems: environment-induced polarization governs protein–ligand binding. To describe the binding energies of protein–ligand complexes more physically, I developed polarization-corrected mechanical embedding (PCME) and built it into PBSA, which keeps the binding-energy calculations fast (to be submitted).

Next *Chemistry on surfaces that rebuild under reaction*
In my Ph.D., I want to bring interfacial water and catalyst restructuring together, and ask how water and reactants behave, and how reactions proceed, on a surface that restructures while the reaction runs. Alongside this, I hope to help advance non-equilibrium electrochemical theory, so that simulations can capture the actual reaction process.

I also want to connect AI agents with robotic platforms to build autonomous labs, where the agent plans and runs both calculations and experiments while the chemist decides which questions matter.